

K2-29b

R-2144

Benchmark run · workflow W8-benchmark-deep · record deep-bench_K2-29_220212 · created 2026-08-12T00:33:04+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459622.6087 +/- 0.0052 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.169 +/- 0.049
Transit depth (Rp/Rs)^2	2.85 +/- 1.67 [%]
Orbital Inclination (inc)	84.29 +/- 2.41
Airmass coefficient 1 (a1)	1.24 +/- 0.49
Airmass coefficient 2 (a2)	-0.15 +/- 0.19
Scatter in the residuals of the lightcurve fit is	40.96 %
Best Comparison Star	None
Optimal Aperture	0.97
Optimal Annulus	3.87
Transit Duration (day)	0.025 +/- 0.022

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.169 ± 0.049 vs 0.1419 (deviation 0.43σ)
Duration fit vs published	0.025 d vs 0.0925 d (deviation 2.38σ)
Mid-time O–C	-10.9 min
Depth SNR	2.7
Residual scatter	40.96 %

Light curve

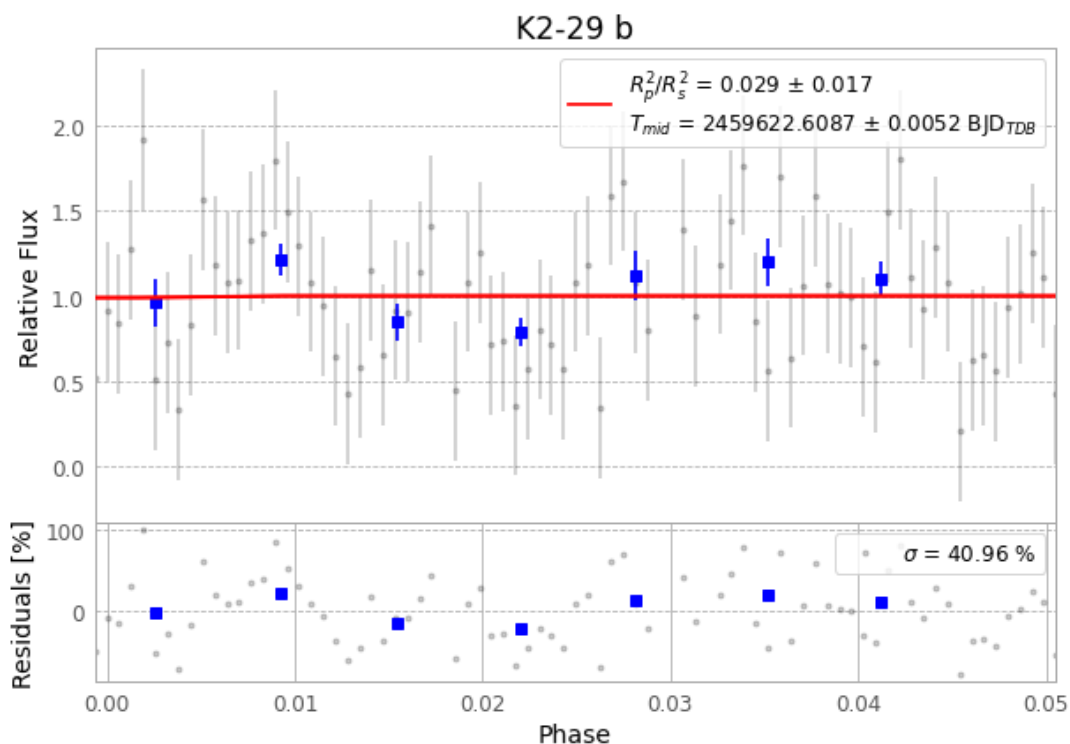


Figure 1 — FinalLightCurve_K2-29 b_2022-02-11.png (SHA-256 33a6618da8562cb1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [215, 297], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 804be083dbb798d2...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

81 FITS frames (53 MB), K2-29220212023012.FITS → K2-29220212063012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest.

Artifacts: bench-K2-29-220212.log (2657637e2519...), FinalLightCurve_K2-29 b_2022-02-11.png (33a6618da856...), FinalLightCurve_K2-29 b_2022-02-11.csv (f3731cdeba52...)

Compute resources

Wall time 146.9 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-12 00:33Z.